

What's in the solar system?

The solar system consists of the Sun, eight planets, three dwarf planets, more than 160 moons, and billions of asteroids and comets. The Sun is by far the most massive member of the system, lying at its center; the other objects all orbit around it. They have existed together since they were formed from a cloud of gas and dust about 4.6 billion years ago.

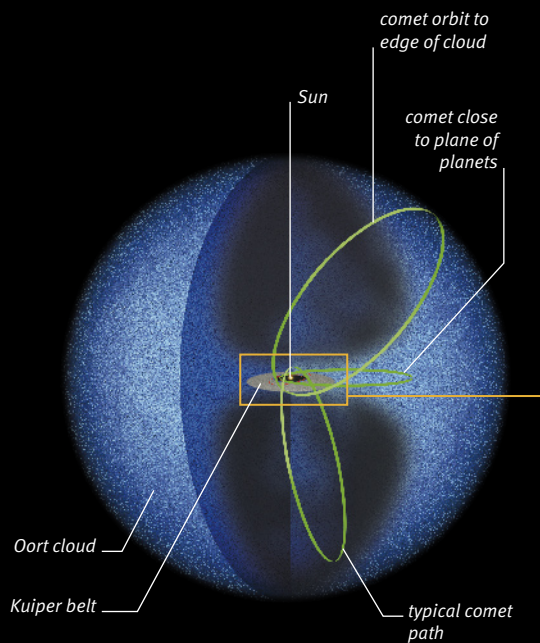
STRUCTURE AND EXTENT

The Sun's immense gravity holds the solar system together. It pulls on all of the other objects in the system and keeps them on their set paths around the Sun. One complete circuit of a path is called an orbit. Each object in the system—from the smallest asteroid, to Jupiter, the largest planet—orbts the Sun.

The orbits taken by the planets, which are the most significant objects after the Sun, are made close to the plane of the Sun's equator. The asteroids that form the main belt, between Mars and Jupiter, also tend to stay close to this plane. Consequently, the

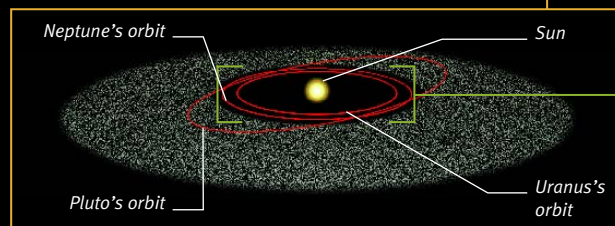
planetary part of the solar system is disk-shaped. The average distance from the Sun to the farthest planet, Neptune, is 2.8 billion miles (4.5 billion km).

Beyond Neptune is the Kuiper belt—a flattened belt of icy, cometlike bodies. And outside of this are the comets themselves. The comets follow randomly inclined orbits that might be close to the planetary plane, above or below the Sun, or anywhere in between. The comets form the Oort cloud, which is a huge spherical cloud surrounding the Kuiper belt and the planetary region. The outer edge of the Oort cloud, which is some 1.6 light-years away and nearly halfway to the closest stars, marks the extent of the Sun's influence. Beyond this lies interstellar space.



Oort cloud

The Oort cloud, which consists of more than a trillion comets, surrounds the planetary region and the Kuiper belt. Three comet orbits are shown here in green: one extends to the edge of the cloud; a second is close to the plane of the planets; the third is a typical elongated orbit.



Kuiper belt

Encircling the planetary region, the Kuiper belt extends from about 3.7 billion miles (6 billion km) from the Sun. It is thought that its outer edge is about 7.4 billion miles (12 billion km) away. The belt is believed to consist of many thousands of objects, more than a thousand of which have been identified.

twice as far from the Sun as Saturn is, Uranus is tilted on its side

Planetary orbits

The planets and asteroids orbit the Sun in a counterclockwise direction as seen from above Earth's north pole. As they orbit, each one also spins. Orbit size and the time taken to complete an orbit increase with distance from the Sun. Orbits are elliptical, which means that a planet's distance from the Sun varies by many millions of miles during its orbit. The planets and their orbits in this image are not shown to scale.

the largest and most massive planet, Jupiter is the one with the fastest spin

liquid water covers more than 70 percent of Earth, which is the only planet known to have life

Mars is the most distant rocky planet from the Sun; it is also the coldest

the hottest and slowest-spinning planet, Venus is similar in size to Earth

Mercury is the smallest and fastest-moving planet; it is also the closest to the Sun

